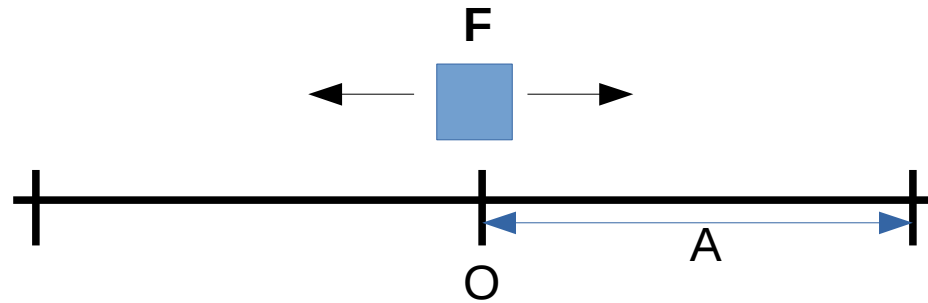


Module 5: Particle dynamics II

Simple harmonic motion



$$a = \frac{d^2 x}{dt^2}$$

$$\frac{d^2 x}{dt^2} + \omega^2 x = 0$$

$$F \propto -x, \quad F = -kx$$

$$a \propto -x, \quad a = -\frac{k}{m}x$$

$$a = -\omega^2 x$$

$$a = \frac{dv}{dt} = \frac{dx}{dt} \frac{dv}{dx}$$

$$a = v \frac{dv}{dx}$$

F = Restoring force

Solve this equation to find the following

1. v
2. x
3. KE
4. PE

Simple harmonic motion

$$a = -\omega^2 x$$

$$a = v \frac{dv}{dx}$$

$$v \frac{dv}{dx} = -\omega^2 x$$

$$\int v dv = \int -\omega^2 x dx$$

$$v^2 = -\omega^2 x^2 + C$$

$$\text{At } x = A, v = 0$$

$$\omega^2 A^2 = C$$

$$v^2 = -\omega^2 x^2 + \omega^2 A^2$$

$$v = \omega \sqrt{A^2 - x^2}$$

$$\frac{v}{\sqrt{A^2 - x^2}} = \omega$$

$$v = \frac{dx}{dt}$$

$$\frac{\frac{dx}{dt}}{\sqrt{A^2 - x^2}} = \omega$$

$$\int \frac{dx}{\sqrt{A^2 - x^2}} = \int \omega dt$$

$$\sin^{-1}\left(\frac{x}{A}\right) = \omega t + \phi$$

$$x = A \sin(\omega t + \phi)$$

Simple harmonic motion

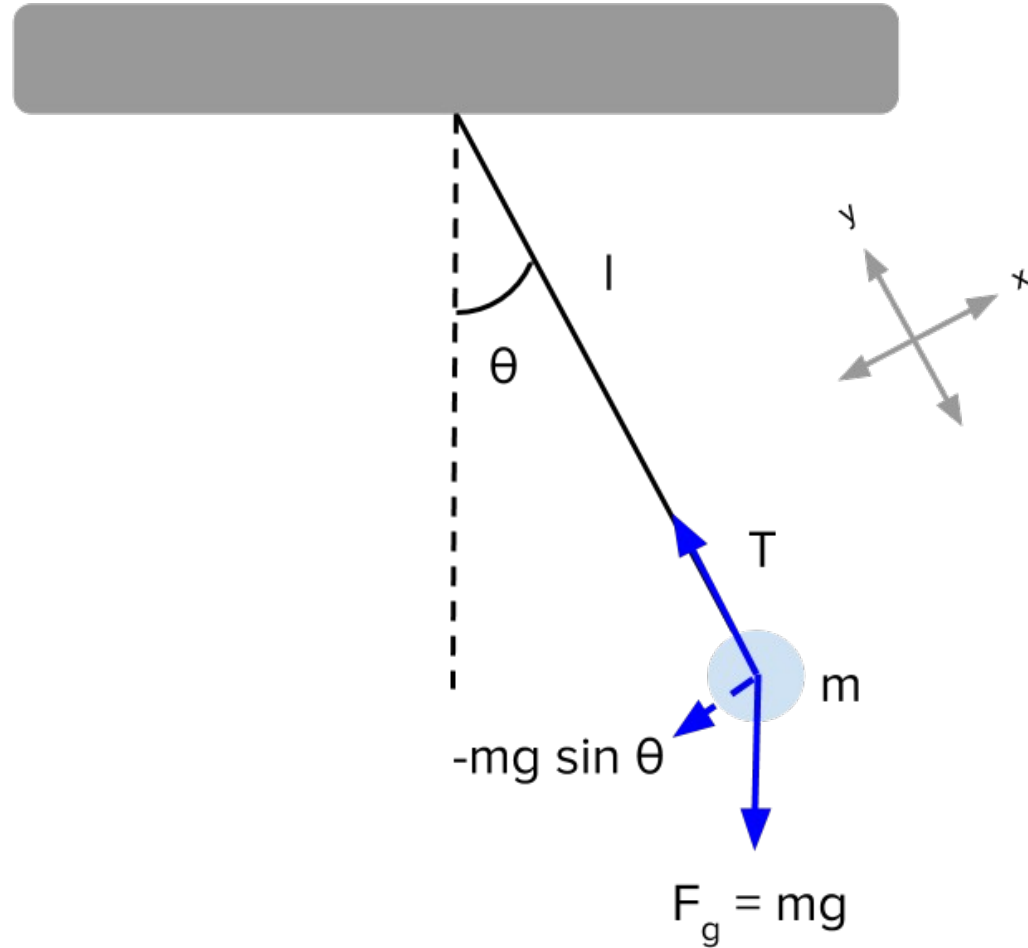
$$v = \omega \sqrt{A^2 - x^2}$$

$$x(t) = A \sin(\omega t + \phi)$$

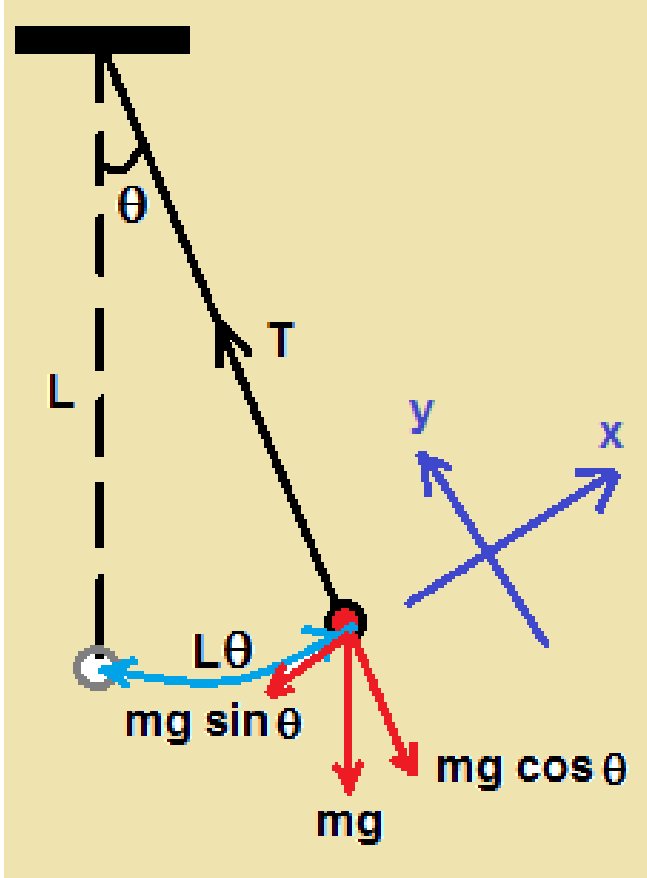
$$KE = \frac{1}{2} m \omega^2 (A^2 - x^2)$$

$$PE = \frac{m \omega^2}{2} A^2 \sin^2(\omega t + \phi)$$

Motion of a simple pendulum



Motion of a simple pendulum



$$F = -mg \sin(\theta)$$

$$a = -g \sin(\theta)$$

$$L \alpha = -g \sin(\theta)$$

$$L \alpha = -g \theta$$

$$L \frac{d^2(\theta)}{dt^2} = -g \theta$$

$$\frac{d^2(\theta)}{dt^2} + \frac{g}{L} \theta = 0$$

Solve this

Damped oscillator

$$m \frac{d^2 \vec{x}}{dt^2} = -k\vec{x} - b \frac{d\vec{x}}{dt}$$

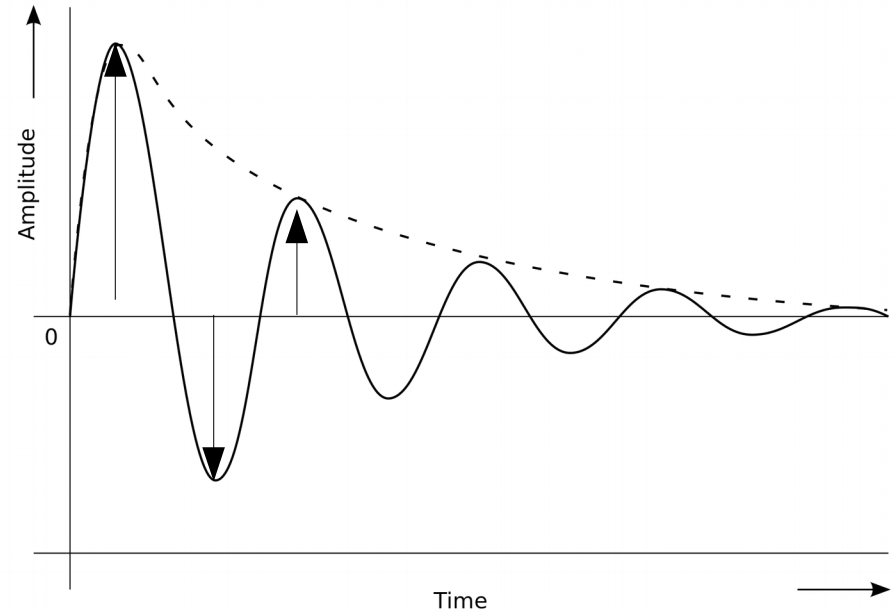
$$m \frac{d^2 \vec{x}}{dt^2} + b \frac{d\vec{x}}{dt} + k\vec{x} = 0$$

$$\frac{d^2 x}{dt^2} + \gamma \frac{dx}{dt} + \omega_0^2 x = 0$$

$$x = A e^{\alpha t}$$

$$\alpha^2 + \gamma \alpha + \omega_0^2 = 0$$

$$\alpha = \frac{-\gamma \pm \sqrt{\gamma^2 - 4\omega_0^2}}{2}$$



Damped oscillator

$$\alpha_1 = \frac{-\gamma + \sqrt{\gamma^2 - 4\omega_0^2}}{2}$$

$$x_1 = A_1 e^{\alpha_1 t}$$

$$\alpha_2 = \frac{-\gamma - \sqrt{\gamma^2 - 4\omega_0^2}}{2}$$

$$x_2 = A_2 e^{\alpha_2 t}$$

So, general solution of this equation is

$$x = A_1 e^{\alpha_1 t} + A_2 e^{\alpha_2 t}$$

Now we can have three cases

(1) $\gamma^2 - 4\omega_0^2 > 0$

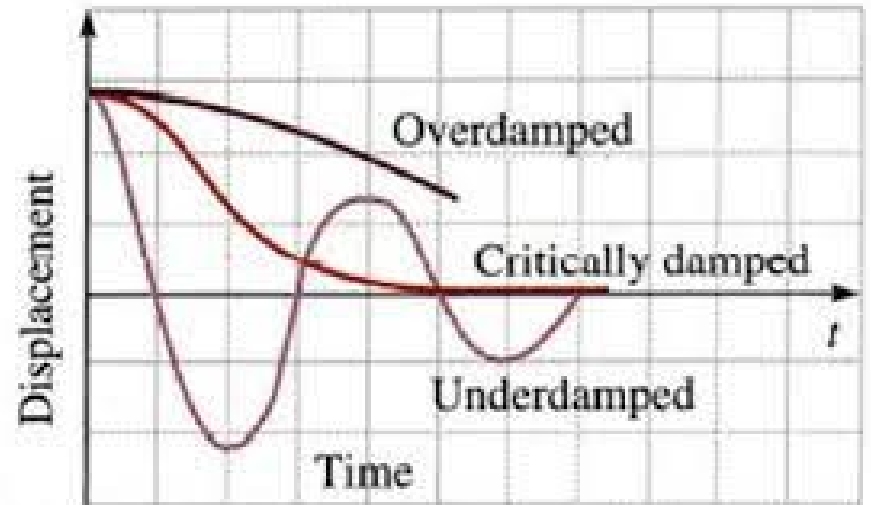
Over-damped

(2) $\gamma^2 - 4\omega_0^2 = 0$

Critically damped

(3) $\gamma^2 - 4\omega_0^2 < 0$

Under-damped



Damped oscillator

For $\gamma^2 - 4\omega_0^2 > 0$

$$\frac{\gamma^2 - 4\omega_0^2}{4} = q^2$$

$$x(t) = A_1 e^{(\frac{-\gamma}{2} + q)t} + A_2 e^{-(\frac{\gamma}{2} + q)t}$$

$$\frac{dx(t)}{dt} = (\frac{-\gamma}{2} + q) A_1 e^{(\frac{-\gamma}{2} + q)t} - (\frac{\gamma}{2} + q) A_2 e^{-(\frac{\gamma}{2} + q)t}$$

For $t=0$, $x=0$, and $\frac{dx}{dt} = v_0$

$$A_1 = -A_2 = \frac{v_0}{2q}$$

$$x(t) = \frac{v_0}{2q} e^{-(\frac{\gamma}{2})t} (e^{qt} - e^{-qt})$$

$$x(t) = \frac{v_0}{2q} e^{\frac{-\gamma t}{2}} \sinh(qt)$$

Damped oscillator

For $\gamma^2 - 4\omega_0^2 = 0$

$$x(t) = A_1 e^{\frac{-\gamma t}{2}} + A_2 e^{\frac{-\gamma t}{2}}$$

$$x(t) = B e^{\frac{-\gamma t}{2}}$$

$$\frac{dx(t)}{dt} = \frac{-\gamma}{2} B e^{\frac{-\gamma t}{2}}$$

This solution fails for $t=0$ at $x=0$, as v_0 becomes zero for this case

Another solution for this equation is $x(t) = Ct e^{\frac{-\gamma t}{2}}$

So, general solution is $x(t) = (B + Ct) e^{\frac{-\gamma t}{2}}$

Damped oscillator

For $\gamma^2 - 4\omega_0^2 < 0$

$$\frac{\sqrt{\gamma^2 - 4\omega_0^2}}{2} = i\omega$$

$$x(t) = e^{\frac{-\gamma t}{2}} (A_1 e^{i\omega t} + A_2 e^{-i\omega t})$$

We know that $e^{ia} = \cos(a) + i\sin(a)$, and $e^{-ia} = \cos(a) - i\sin(a)$

$$x(t) = e^{\frac{-\gamma t}{2}} [(A_1 + A_2)\cos(\omega t) + i(A_1 - A_2)\sin(\omega t)]$$

$$A_1 + A_2 = A\cos(\delta) \quad \text{and} \quad i(A_1 - A_2) = A\sin(\delta)$$

$$x(t) = A e^{\frac{-\gamma t}{2}} [\cos(\omega t)\cos(\delta) + \sin(\omega t)\sin(\delta)]$$

$$x(t) = A e^{\frac{-\gamma t}{2}} \cos(\omega t - \delta)$$

Damped oscillator

For, $t=0$, $x=0$ and $dx/dt = v_0$

$$A \cos(\delta) = 0$$

$$v_0 = -A \left(\frac{\gamma}{2} \cos(\delta) - \omega \sin(\delta) \right)$$

$$\delta = \frac{\pi}{2}, \quad \text{and} \quad A = \frac{v_0}{\omega}$$

$$x(t) = \frac{v_0}{\omega} e^{\frac{-\gamma t}{2}} \sin(\omega t)$$

$$x(t) = A(t) \sin(\omega t)$$

Forced harmonic oscillator

$$m \frac{d^2 x}{dt^2} = -k x - b \frac{dx}{dt} + F_0 \sin(\omega t)$$

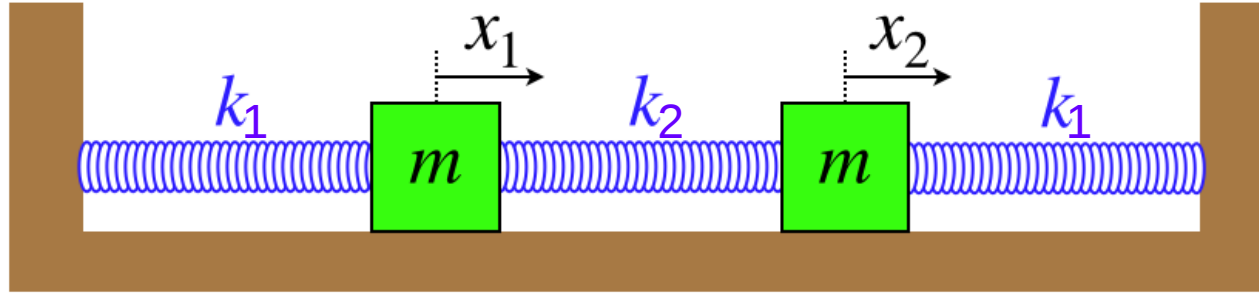
Solution for this equation will look like

$$x(t) = \underbrace{A_h e^{-\gamma t} \sin(\omega_h t + \phi_h)}_{\text{Transient state}} + \underbrace{A \cos(\omega t + \phi)}_{\text{Steady state}}$$

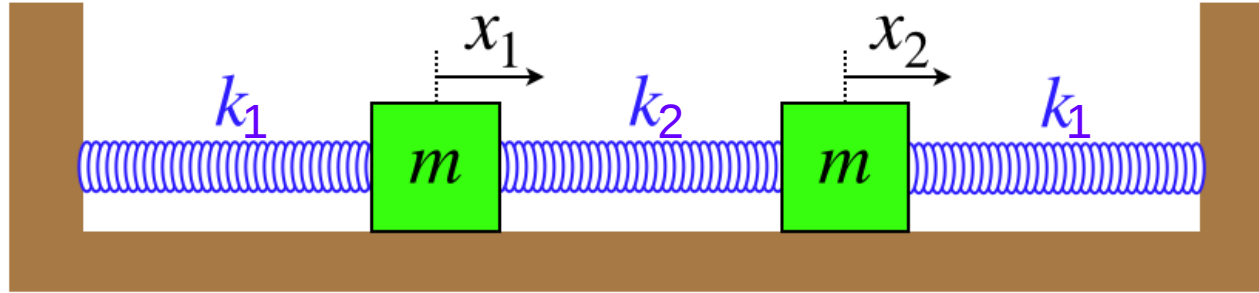
Transient
state

Steady
state

Two body harmonic motion



Two body harmonic motion



$$F_1 = -k_1 x_1 - k_2 (x_1 - x_2)$$
$$m \frac{d^2 x_1}{dt^2} + (k_1 + k_2) x_1 - k_2 x_2 = 0$$
$$\frac{d^2 x_1}{dt^2} + \frac{(k_1 + k_2)}{m} x_1 - \frac{k_2}{m} x_2 = 0$$

A

$$F_2 = -k_1 x_2 - k_2 (x_2 - x_1)$$
$$m \frac{d^2 x_2}{dt^2} + (k_1 + k_2) x_2 - k_2 x_1 = 0$$
$$\frac{d^2 x_2}{dt^2} + \frac{(k_1 + k_2)}{m} x_2 - \frac{k_2}{m} x_1 = 0$$

B

Two body harmonic motion

I. Let us subtract the eq (A) and eq (B)

$$\frac{d^2 x_1}{dt^2} + \frac{(k_1 + k_2)}{m} x_1 - \frac{k_2}{m} x_2 - \frac{d^2 x_2}{dt^2} - \frac{(k_1 + k_2)}{m} x_2 + \frac{k_2}{m} x_1 = 0$$
$$\frac{d^2 (x_1 - x_2)}{dt^2} + \frac{(k_1 + 2k_2)}{m} (x_1 - x_2) = 0$$

Let $q_1(t) = x_1(t) - x_2(t)$ and $(k_1 + 2k_2)/m = \omega_1^2$

$$\frac{d^2 q_1}{dt^2} + \omega_1^2 q_1 = 0$$
$$q_1(t) = A_1 \sin(\omega_1 t + \phi_1)$$

Fast mode

Two body harmonic motion

II. Let us add the eq (A) and eq (B)

$$\frac{d^2 x_1}{dt^2} + \frac{(k_1 + k_2)}{m} x_1 - \frac{k_2}{m} x_2 + \frac{d^2 x_2}{dt^2} + \frac{(k_1 + k_2)}{m} x_2 - \frac{k_2}{m} x_1 = 0$$
$$\frac{d^2 (x_1 + x_2)}{dt^2} + \frac{k_1}{m} (x_1 + x_2) = 0$$

Let $q_2(t) = x_1(t) + x_2(t)$ and $k_1/m = \omega_2^2$

$$\frac{d^2 q_2}{dt^2} + \omega_2^2 q_2 = 0$$
$$q_2(t) = A_2 \sin(\omega_2 t + \phi_2)$$

Slow mode

Two body harmonic motion

So, our general solution will be

$$x_1(t) = \frac{1}{2}(q_1 + q_2)$$

$$x_1(t) = \frac{1}{2}(A_1 \sin(\omega_1 t + \phi_1) + A_2 \sin(\omega_2 t + \phi_2))$$

$$x_2(t) = \frac{1}{2}(q_2 - q_1)$$

$$x_2(t) = \frac{1}{2}(A_2 \sin(\omega_2 t + \phi_2) - A_1 \sin(\omega_1 t + \phi_1))$$